

National University of Computer and Emerging Sciences



Lab Manual 13

Programming Fundamentals

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Semester	Fall 2025

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Activities:

Note: Include input validation in programs wherever applicable. Don't use any build-in function.

1. Write a C++ program that manages a small Library Inventory System using file handling.
The program should maintain a file "library.txt" that stores book records in the following format:
BookID Title Author Copies

Functional Requirements

Your program must provide the following features using separate functions:

Add New Books

- Create/overwrite the file.
- Ask the user how many books to enter.
- For each book input:
 - Book ID (integer)
 - Title (single word for simplicity)
 - Author (single word)
 - Number of Copies (integer)

Display All Books

- Read all book records from library.txt
- Display them in proper formatted output.

Append More Books

- Allow the user to add new books to the existing file.

Search for a Book by Book ID

- Read the file and search for a specific Book ID.
- Display the book details if found, otherwise show "Not Found".

File Format Example (library.txt)

```
101 CppPrimer Lippman 4
102 Algorithms Sedgewick 3
103 DBMS Navathe 5
```

2. Write a C++ program that manages a Vehicle Service Center using file handling.
The program should store service records in a file named "service.txt" in the following format:
ServiceID VehicleNo ServiceType Cost

Functional Requirements

Your program must implement the following features using separate functions:

Create Service Records

- Overwrite or create service.txt.
- Ask the user how many service entries to enter.
- For each record input:
 - Service ID (integer)
 - Vehicle Number (e.g., ABC123)
 - Service Type (single word, e.g., OilChange, Tuning)
 - Service Cost (float/int)

Display All Service Records (Read Mode)

- Read all entries from the file.
- Display them in formatted form:

Append New Service Records

- Add new entries.

Calculate Total Revenue Generated

- Read the file.
- Sum all service costs.

File Format Example (service.txt)

101 ABC123 OilChange 1500

102 DEF987 Tuning 2500

103 GHI555 WheelAlign 1800